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Project Group 07: Yelp - Data Insights: Building a Business Recommender System

Background

Online platforms have grown rapidly due to user-centered designs that make products and services more accessible through their user-friendly interfaces. One component of the user-centered design has been recommender systems which leverage big data to analyze users' interactions, similar user experiences, and overall reviews. These recommender systems enhance user engagement and drive possible user retention which in turn helps to create revenue growth. Platforms are always looking for ways to enhance these algorithms to better understand how the user behaves. The goal of this project is to optimize the recommendation algorithm to help create a more personalized user experience thus generating more revenue and growth for the platforms.

Problem Definition

The project aims to solve the problem of connecting users with businesses that match their preferences by analyzing user interaction data and business locations. Inputs for the system include user interaction data, business details, and geographic coordinates while the output is a personalized list of recommended businesses for each user based on user preferences, interactions, and proximity.

Motivation

Users often face challenges in navigating the overwhelming number of options available on platforms, making it difficult to discover businesses or services that align with their preferences. Recommender systems address this by personalizing suggestions based on user interactions and other factors, enhancing user satisfaction and engagement while improving business outcomes. ML techniques excel at identifying patterns in large, complex datasets and dynamically adapting to new data, making them ideal for evolving platforms. The dataset is *Yelp Complete Open Dataset* (<https://www.kaggle.com/datasets/adamamer2001/yelp-complete-open-dataset-2024>) obtained from Kaggle comprising six rich data files totaling over 9 GB, including business details, user interactions, reviews, and engagement metrics.

Literature Review

Recommender systems have become an essential feature on most online platforms, enhancing user experience by offering personalized suggestions for products or services. Many researchers have addressed this problem using various approaches. Collaborative filtering, introduced by Yifan Hu et al. (2008), learns user-item preferences by analyzing user-item interactions and is highly effective for large datasets in e-commerce and social media. More recently a paper called

Relational Collaborative Topic Regression for Recommender Systems et al. Wang (2015) that uses networks with collaborative filtering to enhance these recommendations. We aim to build on these foundational approaches, leveraging collaborative filtering to analyze user-business interactions while integrating content-based methods to utilize business attributes. This combination aligns with research trends and is well-suited for the structure and scale of the Yelp dataset we are using.

Approach

We plan to combine collaborative filtering with content-based methods, using ML algorithms like KNN or Random Forest to enhance accuracy. Using other methods like cosine-wise similarity, Jaccard similarity, and collaborative filtering will also leverage the Yelp dataset's user-business interactions, while content-based methods will incorporate location data, such as population density and distances between businesses, to enhance recommendation accuracy. To evaluate the success of our model, we will use metrics such as accuracy, precision, recall, and F1 score, ensuring a comprehensive assessment of its performance in generating relevant and personalized business recommendations. This approach balances user interaction data with contextual factors like location, enabling us to provide meaningful insights and accurate predictions.

Thankachan, Karun. "The Best Research Papers in Recommender Systems".
Feb 26, 2023.
<https://medium.com/codex/the-best-research-papers-in-recommender-systems-335dbba3b252>